ECS 1100 · SECTOR 4

Energy Science & Technology

How many majors does it take to keep the power flowing?

START HERE

Who keeps the lights on?

What has to work together before your phone can charge tonight?



Illustration: electric grid at dusk (schematic).

What is Energy Science & Technology?

A field that connects the physical science of energy with useful, reliable technology.

ENERGY SCIENCE

Understand energy sources, conversions, materials and system behavior.

ENERGY TECHNOLOGY

Generate	Solar, wind, nuclear and thermal plants
Convert & store	Motors, inverters, batteries and heat
Deliver	Transmission lines and substations
Manage	Sensors, forecasts, controls and markets
Use efficiently	Buildings, data centers and vehicles

Sources: EIA Electricity explained; DOE Storage.

From an energy source to everyday use

Different stages require different technologies — and different engineering skills.

STAGE 1

Generate

Turn primary energy into electricity.

Examples: solar PV, wind, gas turbines, nuclear plants.

EE · ME · MSE

STAGE 2

Convert & store

Change voltage or electrical form; store energy for later.

Examples: inverters, transformers, batteries.

EE · CE · MSE · ME

STAGE 3

Move power

Transport and deliver electricity over interconnected networks.

Examples: transmission, substations, distribution.

EE · SysE · CE

STAGE 4

Manage & use

Balance supply and demand and use energy productively.

Examples: software, controls, HVAC, data centers.

CS · SE · DS · ME · SysE

ACROSS EVERY STAGE: safety, reliability, cost, environmental impacts, and communication.

Sources: EIA Delivery to consumers; DOE Storage.

Energy is more than solar and wind

The industry combines many primary energy sources with equipment, storage and efficient use.

SOLAR

Sunlight → electricity

PV cells, inverters, materials

WIND

Moving air → electricity

Blades, generators, controls

NATURAL GAS

Fuel → heat → electricity

Turbines, thermal systems

NUCLEAR

Fission heat → electricity

Reactor systems, steam cycles

HYDRO

Moving water → electricity

Turbines, civil + power systems

GEOTHERMAL

Earth heat → useful energy

Thermal cycles, drilling, pumps

Source: EIA Electricity explained; DOE Storage. Examples are illustrative.

One power system, many majors

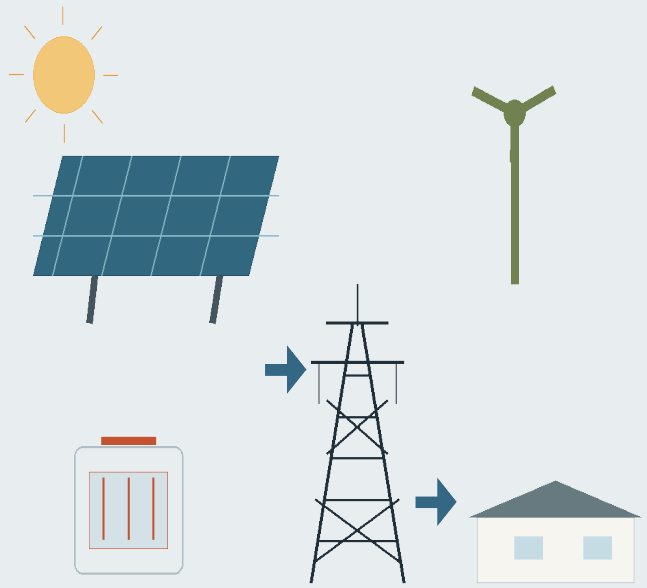
Systems (SysE)
Brings generation, storage and controls together.

Electrical (EE)
Designs power flow, converters and protection.

Mechanical (ME)
Handles turbines, structures and cooling.

Materials (MSE)
Improves batteries and energy materials.

GENERATION → GRID → USE



Storage Delivery Demand

Computer (CE)
Sensors, firmware and embedded power control.

CS + Software (SE)
Digital tools, control platforms, secure apps.

Data Science (DS)
Forecasts demand and detects anomalies.

Human factors (BME)*
Worker safety and ergonomics: indirect pathway.

* BME is an interdisciplinary connection here, not a core power-engineering hiring route.

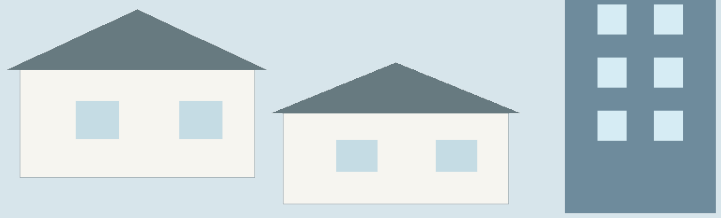
Conceptual major-to-system mapping; sources: EIA, DOE, job-posting log.

Start with people: a hot Texas afternoon

Behind a demand curve are homes, hospitals, businesses and students trying to work and stay cool.

SUMMER PEAK

Cooling + care + work



Illustrative community load — no actual demand data shown.

THE HUMAN NEED

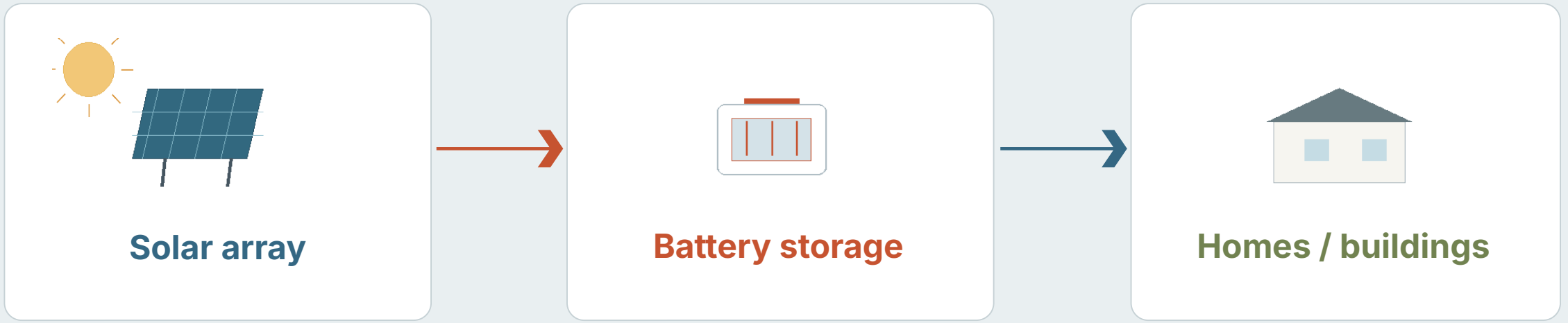
Keep essential services powered while demand changes through the day.

1. **Supply** — Enough usable electricity needs to be available.
2. **Delivery** — Lines and equipment must handle local demand.
3. **Decisions** — Forecasting, storage and demand response can help.

Conceptual case built from EIA electricity use and DOE storage concepts.

One solution: solar + battery + smart controls

Solar can serve loads directly; a battery can store surplus energy for later use.



SMART CONTROLS — **When to charge? When to discharge? How much power is needed?**

DAY

Solar can power loads and charge batteries.

EVENING

Stored energy can be used later.

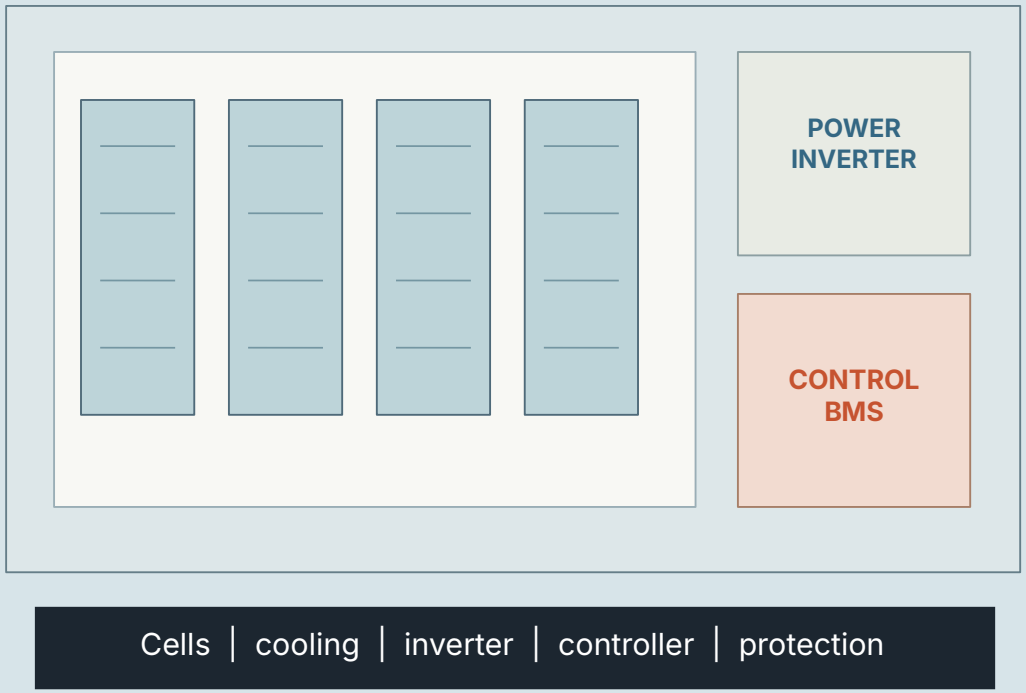
ANYTIME

Grid connections and system limits still matter.

Sources: DOE Onsite energy technologies; DOE Storage. Example is illustrative.

Inside a battery energy storage system

One container needs materials, cooling, electronics, embedded control and software to work safely.



Illustrative cutaway: simplified, not to scale.

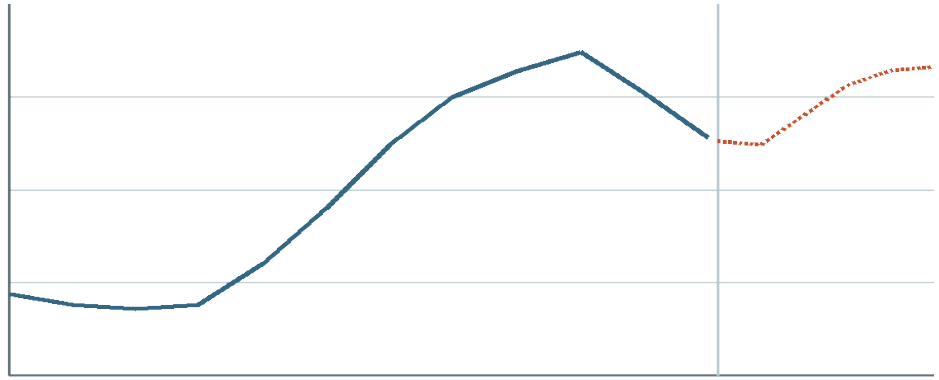
MSE	Design battery materials and cells.
ME	Keep cells within safe temperatures.
EE	Convert and protect electrical power.
CE	Monitor cells with sensors and firmware.
SysE · CS · SE · DS	Integrate, monitor, forecast and optimize.

Sources: DOE Energy Storage Safety; DOE battery applications.

Digital energy: turning data into decisions

Computer Science, Software Engineering, and Data Science also operate inside physical energy systems.

ILLUSTRATIVE ELECTRICITY DEMAND



Observed

Forecast →

Conceptual curve. Not actual ERCOT data.

1 **Measure**
Meters and sensors collect information.

2 **Forecast**
Models estimate future demand or generation.

3 **Decide**
Control systems schedule resources.

4 **Verify**
Engineers monitor performance and errors.

Sources: DOE onsite storage controls; UTD REVT digital twins and control research.

Reliability case: the Texas winter storm

February 2021: multiple parts of the energy system experienced failures at the same time.

1 Extreme cold

Freezing weather challenged equipment and fuel supply.

2 Generation becomes unavailable

Some plants tripped, lost output or could not start.

3 Emergency grid actions

Demand exceeded available supply; grid operators ordered load shedding.

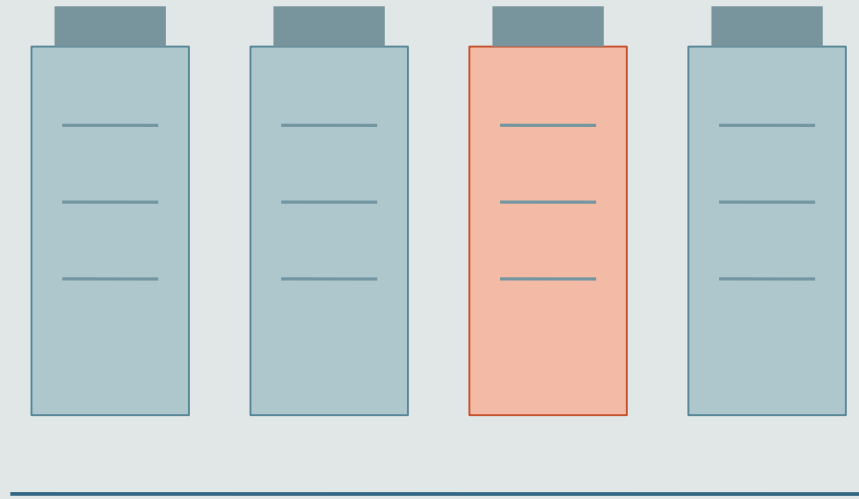
ENGINEERING LESSON

Weatherproofing, fuel supply, power equipment and grid operations must be designed as an interconnected reliability problem.

Source: FERC/NERC Final Report (Nov. 16, 2021); multiple fuel types were affected.

Safety case: batteries must manage heat

A storage system is designed to reduce both the likelihood and consequences of failure.



Cell temperatures, insulation and spacing are design constraints.

1 Prevent

Materials, electrical protection and controlled charging.

2 Detect

Temperature sensing, monitoring and fault response.

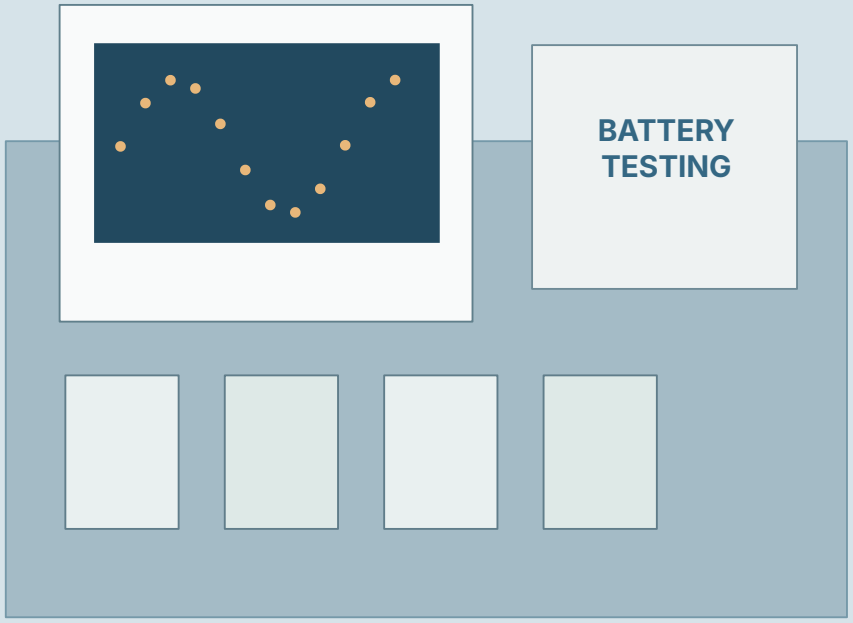
3 Contain & prepare

Thermal management, system testing, incident planning.

Source: DOE Energy Storage Safety Strategic Plan (2024). Risk is not an inevitable outcome.

Research turns energy questions into technology

At UT Dallas, energy research spans converters, electric machines, battery chemistry and manufacturing.



Schematic illustration — not a photo of a particular UTD lab.

1

REVT / Power electronics

Converter reliability, electric drives and grid-related control.

2

MSE / Battery materials

Electrolytes, interfaces and next-generation rechargeable cells.

3

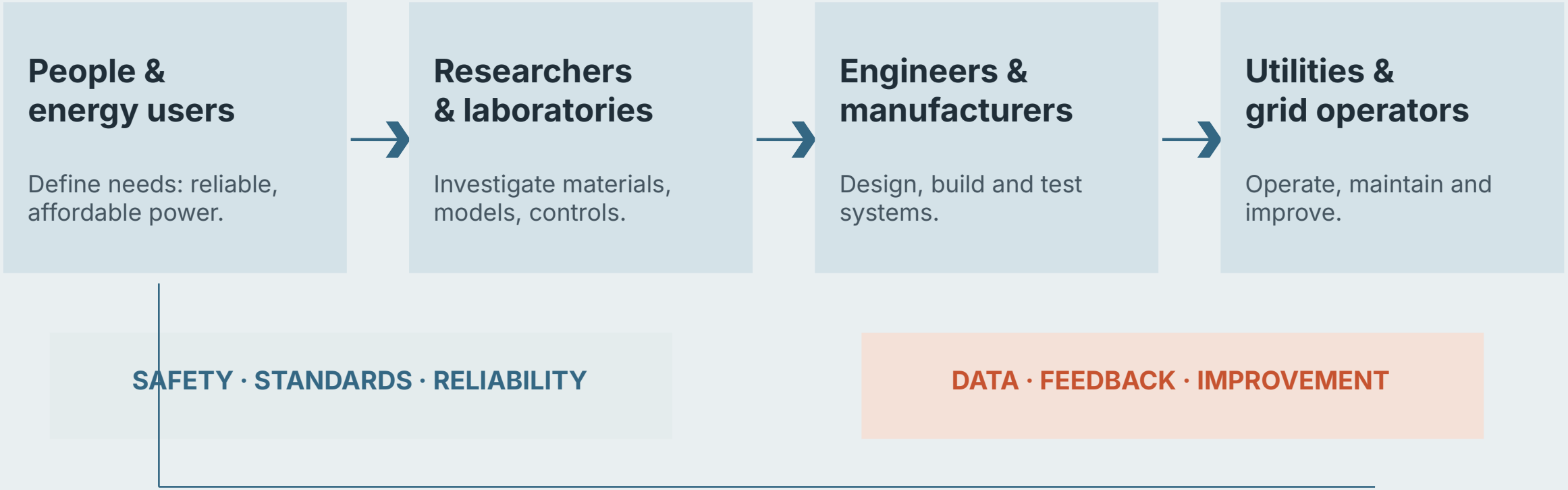
BEACONS / Prototyping

Moves battery ideas toward testing, manufacturing and scale-up.

Sources: UT Dallas REVT; MSE battery research; UT Dallas BEACONS (accessed Oct. 22, 2026).

How energy innovation works

From community needs to reliable energy technology — with feedback at every step.



The next design begins with what the last system taught us.

Conceptual innovation cycle; sources: UTD REVT, BEACONS; EIA, DOE.

Engineering in energy is not just about building more power plants.

It is about
meeting people’s needs,
keeping systems safe and reliable,
and solving problems across disciplines.

Real postings across the energy system

Examples from the Sector 4 job log, captured Sept. 22, 2026. Posting status may have changed.

GRID & POWER

Move and protect electricity

Siemens

Power Systems Technical Consultant Intern

Internship · EE · Austin

Paragon Energy Solutions

Electrical Engineer I

Entry level · EE · Fort Worth

STORAGE & HARDWARE

Build energy products

Sol-Ark

Electrical Systems Test Engineer

CE / EE · Allen

Tesla

Manufacturing Engineer, Battery Manufacturing

MSE / manufacturing · Austin

BUILDINGS & EFFICIENCY

Use energy well

Trane Technologies

Entry Level Energy Engineer

Entry level · ME · San Antonio

Leidos

Energy Engineer

ME · Dallas

DIGITAL & MARKETS

Use data to make decisions

Enchanted Rock

Software Engineer, C#/.NET

SE / CS · Houston

Macquarie Group

Junior Energy Quantitative Analyst

Junior · DS · Houston

Source: ECS1100 Sector 4 Job Log (collected Sep. 22, 2026); confirm links before applying.

What the postings ask for

Skills in physical energy systems: learning paths for EE, CE, ME, and MSE.

Electrical Engineering

- Power and circuit fundamentals
- Electrical design, testing and protection
- Substations and distribution

Computer Engineering

- Embedded hardware and firmware
- Sensors, diagnostics and controls
- Battery management and SCADA

Mechanical Engineering

- Thermal and fluid systems
- HVAC, cooling and mechanical design
- Manufacturing and commissioning

Materials Science

- Battery chemistry and materials
- Materials testing and process design
- Quality, recycling and manufacturing

SHARED EXPECTATIONS: problem-solving · teamwork · testing · communication · documentation

Recurring themes in the Sector 4 job log, collected Sep. 22, 2026. Exact requirements vary.

What the postings ask for

Digital systems and interdisciplinary connections in the energy sector.

Systems Engineering

- Requirements and interfaces
- Integration and verification
- Reliability, risk and planning

Software Engineering

- Backend systems and APIs
- Testing, deployment, version control
- Monitoring and energy platforms

Computer Science

- Programming and algorithms
- Automation, controls and AI
- Operational software and security

Data Science

- Python / SQL and statistics
- Forecasting and time-series models
- Analysis, dashboards and decisions

Biomedical Engineering: indirect routes via worker health, industrial hygiene, ergonomics and human factors.

First-year takeaway: pick a problem, then look for the majors, skills and projects behind it.

Recurring themes in Sector 4 job log (Sep. 22, 2026). BME is an indirect pathway.

What energy employers ask for, by major

From 29 real internship and entry-level postings, Oct 2026 (15 EE · 14 CE · 14 ME; some roles take more than one major)

EE Electrical Engineering

Power systems & grid basics

Three-phase circuits, power flow, protection, transformers

Seen at: Oncor, SEL, Hitachi, GE Vernova

Modeling & simulation

MATLAB/Simulink, PSCAD, Python, SQL

Seen at: Oncor, SEL, GE Vernova

Hands-on test & debug

Relays, oscilloscopes, CAN bus, failure investigation

Seen at: SEL, Tesla Energy, ExxonMobil

Safety, codes & drawings

Regulations, schematics, shop drawings, AutoCAD

Seen at: Oncor, Mortenson, Vistra

CE Computer Engineering

Programming

Python most often; also Go, Java, C/C++, C#, SQL

Seen at: Tesla, Chevron, GE Vernova

Embedded, firmware & test

CAN, Linux logs, test automation (Robot Framework, LabVIEW)

Seen at: Tesla Energy, GE Vernova, Base Power

Cloud & DevOps

Azure/AWS, Git, CI/CD, Kubernetes

Seen at: Chevron, HF Sinclair, GE Vernova

Data & AI for operations

Data historians, machine learning, AI-assisted coding

Seen at: Energy Transfer, Halliburton, Chevron

ME Mechanical Engineering

Thermal-fluids & turbines

Rotating equipment, pipe/vessel sizing, flow measurement

Seen at: Vistra, Mitsubishi, Energy Transfer

Design tools

CAD (SolidWorks, AutoCAD), GD&T, ANSYS, MATLAB

Seen at: Kairos Power, Mitsubishi Power

Root cause & reliability

Failure analysis, FMEA, inspections

Seen at: Vistra, Mitsubishi, Kairos, ExxonMobil

Manufacturing & improvement

Lean Six Sigma, plant trials, prototyping

Seen at: GE Vernova, Kairos Power, ExxonMobil

What almost every posting asks for

Across all 29 postings, regardless of major

20 of 29

Communication
Write reports, document test results, present to a team or leadership

13 of 29

Teamwork
Cross-functional teams: operations, maintenance, IT, field crews

10 of 29

Grades
GPA of 3.0+ (or a UK 2:1) stated, or good academic standing

9 of 29

Python
Named in EE, CE and ME postings alike

Read the fine print

- **13 of 29** require US work authorization without sponsorship; Vistra excludes OPT/CPT

13 of 29 require onsite work, relocation, field travel or a driver's license

Many internships say prior projects or research are **preferred, not required**

Your move this year: build one project in one skill from your column, and practice explaining it in 60 seconds.